therapies of nervous and mental diseases, new generation of neuromorphic energy-saving computing systems, artificial brain, artificial intellect and neurorobotics.[1–4]

The goals of this work are to describe an alternative information-communicative theory of organization of the brain and the information principles of its operation, to demonstrate its scientific and practical meaning for neurology, neurosciences, and brain modeling, and to explain the mechanisms of formation and loss of reflexes, consciousness, memories, and intellect in the disordered information structure of brain.

Conventional scientific concepts of brain organization and structure can be summarized as follows:

1. The brain is a multilevel, multifunctional structure in the central nervous system (CNS) of mammalian (including human) organisms intended to receive, transfer, process, and store information received from the body's organs, systems, and environment. The CNS regulates and maintains homeostasis in the organism, promoting its survival and adaptation under constantly changing conditions and extreme environmental factors.[2,3,5–8]
2. A number of systemic morphofunctional levels of the brain and CNS can be distinguished, ie, the genome, transcriptome, proteome, and metabolome levels, synaptic level, level of the cell, level of tissue circuits, level of brain regions, level of connectivity, level of the whole brain, and the cognitive level.[1]
3. Nerve cells (neurons) are the basic informational structural and functional units of the brain that, being connected through multiple synapses, form neural circuits[2] providing a morphologic substrate for functional systems in the brain.[9,10] Organized into neural circuits, neurons have strict topical localization and functional specialization (eg, motor, visual, proprioceptive) in the CNS and cannot be restored if injured.[4]
4. Inside the neuron, information is transmitted by electrical package-impulse transfer through the nodes of Ranvier on its axon, while between the neurons information is transmitted chemically through neuromediators, growth factors, and neurotransmitters.[6,11,12] Anatomic structures connecting neurons are called synapses and provide a morphological substrate for formation of neural networks.[1,2]
5. Neural circuits are the basic information structures of the brain, and consist of neurons and synapses that support the basic functions of life in a human/mammal, and record and store memory, cognition, and intellectual-mnestic activity.[13,14] The cerebral cortex can be perceived as an information system consisting of hierarchical groups of neural circuits.[15]
6. Overall, the brain is a multilevel neural network[16,17] that receives, processes, and stores information which is evenly distributed among all neurons of the brain due to the holographic principle.[10,18] At the same time, the human brain can also be represented as a hologram capable of forming different functional systems to satisfy the needs of the organism and to achieve a useful adaptive result.[9,10]
7. Memory is one of the basic functions of the brain, is evenly distributed among all cells of brain cortex and subcortical nodes, and is stored in the protein and genome networks of neurons,[19] permitting recording, storage, and reproduction of the information in every region of this multilayered neural network.[20,21] The main substrate of the brain responsible for memories is the hippocampus, activation of which by optogenetically manipulating memory engram-bearing cells can result in false memories.[22]
8. Morphofunctional systems in the brain are based on circuits providing functional cortical neurodynamic integration[23–25] of different brain regions and brain formations and this is manifested in bioelectrical activity, oscillatory processes,[26] and magnetoencephalographic manifestations of brain functioning.
9. Neurodynamic integration forms cognitive functions of neural circuits of brain cortex[27] based on the principles of free energy.[28–30] Perception and thinking are also the circuit functions; it is the cooperation of different brain areas that keep constantly adapting on the basis of the solved tasks, internal resources of the brain and biological limitations.[31]
10. Information is processed in the brain cortex at the level of neural codes[32–34] by activation of the membrane currents of the cortex, as well as simultaneous interaction of different cell levels of the cortex[35] which is manifested in cognitive functions. Quantum mechanisms of consciousness formation[36–38] and neuroconnectivity of cerebral cortex underlie the information processing.[39,40]